GLOW Blend (GHK-Cu / BPC-157 / TB-500 (TB4))

1.4 mg daily for skin, hair, healthspan; 2.8 mg daily for tissue repair

GLOW is an experimental peptide blend proposed to support skin quality, wound repair, and connective-tissue resilience through plausible complementary actions of its components. [1–3,8,13] The common formulation discussed here contains GHK-Cu, BPC-157, and full-length thymosin β4 (TB4) in a 5:1:1 mass ratio. GHK-Cu influences collagen and extracellular-matrix production in cultured fibroblasts and experimental wounds. [2,3] BPC-157 has tendon-repair and vascular-signaling findings in preclinical models. [5,13,31] TB4 regulates actin dynamics and has wound-healing, cell-migration, and angiogenic effects in experimental systems. [8,9,17] Synergy within GLOW is plausible. [1,8,13,31]

💉 💉 Probably safe. [18–21,32,34]

⭐ ⭐ Plausibly effective for selected repair and cosmetic applications, based mainly on component research. [2,3,8,13,14,22]

C Angiogenic — precaution around active malignancy or elevated cancer risk, based on experimental tumor biology. [25,26]

Synthetic TB4 was generally well tolerated during single IV exposure and 14 days of daily IV administration in a placebo-controlled study of 40 healthy volunteers. [18] In a BPC-157 pilot no side effects or changes in the measured cardiac, hepatic, renal, thyroid, or glucose markers were reported. [32] GHK-Cu's animal findings and topical human experience provide indirect support for plausible tolerability. [14,38] A small, short BPC-157 pilot had unchanged measured liver, kidney, thyroid, and glucose markers. [32] The TB4 study reported no serious adverse events or dose-limiting toxicity. [18]

Individual components improved wound or tendon repair in specific animal models, including TB4 in rat skin wounds and BPC-157 after rat Achilles-tendon transection. [8,13] GHK-Cu supported extracellular-matrix production and remodeling in cell and wound models. [2,3] GHK-Cu increased collagen synthesis in cultured fibroblasts and altered matrix production in rat wounds. [2,3] TB4 increased hair growth in rats and mice and affected follicular-cell migration and differentiation. [22] Corresponding human repair effects are plausible.

Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.

The names TB4, Tβ4, and TB-500 are used inconsistently in the research-use-only market. Full-length TB4 and the acetylated 17–23 fragment, Ac-LKKTETQ, are different molecules; a vendor name alone does not establish a sample's identity. [6,29,35]

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Janoshik distinguishes the full 43-amino-acid chain from fragments in its analysis. "TB-500 (TB-4)" denotes the entire peptide, "TB-500 (Frag)" denotes the 17–23 fragment, and "TB-500" alone denotes a fragment whose identity is specified in the report comments. [29]

This survey concerns GLOW containing full-length, 43-amino-acid TB4. "TB-500 (TB4)" below denotes that form.

Dosage & Patterns

These speculative, experimental daily patterns often discussed:

- **1.4 mg daily GLOW for skin, hair, healthcare**
  - ~ 1 mg GHK-CU / daily
  - ~ 200 mcg BPC-157 / daily
  - ~ 200 mcg TB-500 (TB4) / daily
- **2.8 mg daily GLOW for tissue repair**
  - ~ 2 mg GHK-CU / daily
  - ~ 400 mcg BPC-157 / daily
  - ~ 400 mcg TB-500 (TB4) / daily

The lower pattern is proposed for cosmetic goals and healthspan; the higher pattern for tissue repair.

Benefits

- Plausibly smoother, healthier-looking skin, including possible changes in hydration, elasticity, and tone. [1–3,14]
- Plausible human wound-repair. [8]
- Faster recovery from soft-tissue strains is plausible through effects on cell migration, vascular repair, and tissue remodeling. [8,11,17]
- Possible reduction in joint discomfort. [33]
- Plausible support for recovery from minor soft-tissue injuries and overuse irritation through effects on vascular repair and extracellular-matrix remodeling. Relief of delayed-onset muscle soreness remains a community-reported benefit. [3,8,11,31]
- Calmer old injury sites and fewer perceived flare-ups are anecdotal possibilities; the proposed repair rationale draws on component wound and tendon findings. [8,13]
- Human tendon recovery and improved resilience during training are plausible. [5,8,13]
- Improved human microvascular function, nutrient delivery, and tissue oxygenation under mechanical load are plausible extrapolations. [9,11]
- More even texture, fewer fine lines, and improved firmness over several weeks are plausible cosmetic goals. [1,14]
- Generalized tissue comfort, less tightness, fewer aches, and smoother movement are community-reported possibilities, distinct from the measured outcomes of component studies.

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Applications

- Chronic soft-tissue irritation. [8,13]
- Older injuries that flare under load, speculative application.
- High training volume, speculative.
- Age-related changes in skin quality. [1,14]
- Demanding physical jobs or lifestyles, speculative.
- Recovery from minor strains or overuse. [8,13]
- Low-grade systemic inflammation, speculative. [7,38]

Cautions

- **Copper metabolism disorders:** Impaired copper handling, particularly Wilson disease, warrants caution with a copper-containing compound. [37]
- **Liver disease:** TB4's fibrosis biology is context-dependent. Exogenous TB4 reduced chemically induced liver injury and fibrosis in rodents, whereas deleting endogenous TB4 in activated hepatic stellate cells also reduced fibrosis; restoring its expression reversed that benefit. These different interventions should not be treated as equivalent. [23,24]
- **Active cancer or elevated cancer risk:** Experimental tumor-cell studies justify precaution; the direction and magnitude of clinical risk depend on tumor biology and exposure. [25,26]

Drivers of Diminished Response and Mitigation Strategies

These peptides have different mechanisms, including TB4's actin binding and BPC-157-associated VEGFR2 signaling; a blanket claim that none interacts with receptor pathways is inappropriate. [17,31] Adaptation within repair pathways is plausible.

GLOW: Plausible Healthspan Rationale

GLOW's proposed healthspan rationale combines GHK-related matrix and gene-expression effects, TB4's cytoskeletal and vascular-repair biology, and BPC-157's gastrointestinal and endothelial findings. [1–3,8,17,31,40] Complementary support for tissue integrity and repair is plausible; a broad functional healthspan benefit remains speculative.

GHK-related gene-expression studies suggest possible effects on repair, inflammatory, and antioxidant programs in cultured human cell systems. [30,39] Supporting collagen, elastin, glycosaminoglycan, and metalloprotein regulation is a plausible route to maintaining connective-tissue resilience during aging. [1–3]

TB-500 (TB4) contributes actin regulation, cell migration, and vascular-repair mechanisms to the healthspan rationale. [9,11,17] Complementarity with GHK-Cu's matrix effects remains plausible. [2,3]

A hypothesis linking BPC-157's intestinal-barrier preservation and vascular repair to lower chronic inflammatory burden, improved nutrient delivery, and tissue resilience is biologically

plausible. [4,31,40]. Extending that hypothesis to broad human healthspan benefit remains speculative.

Synergies in combining these three peptides are plausible, providing a rationale for supporting tissue resilience and healthspan. [1-3,8,13,31]

GLOW and Cancer Risk

The cancer concern involves possible support of an existing malignancy through migration and angiogenesis. Increasing TB4 expression within melanoma cells promoted tumor growth and metastasis in mice. [25]

Exposure Alternatives

Three proposed patterns change component exposure; their comparative cancer risk is speculative:

- GHK-Cu alone retains a plausible matrix-repair rationale while removing TB4 and BPC-157 exposure. [2,3]
- GHK-Cu with BPC-157 retains a plausible soft-tissue-repair rationale while omitting TB4. This change does not establish a lower clinical cancer risk. [2,3,13,25,26]
- Alternating GLOW and GHK-Cu is a speculative approach to reducing continuous TB4 and BPC-157 exposure.

Zinc Supplementation

SC GLOW introduces copper directly into systemic circulation, potentially altering copper–zinc homeostasis. Intracellular copper binds strongly to metallothioneins, cysteine-rich proteins involved in metal sequestration and regulation. Copper has a higher affinity for metallothionein than zinc and can displace bound zinc, potentially altering its intracellular distribution and availability for zinc-dependent enzymes, transcription factors, and signaling pathways. Sustained copper exposure could therefore produce functional zinc insufficiency through altered metal sequestration. Unlike the well-established reduction in copper absorption caused by excessive oral zinc, clinically significant zinc depletion from subcutaneous GHK-Cu remains a mechanistically plausible but unproven effect in humans.

Adult zinc requirements are 8 mg daily for nonpregnant, nonlactating women and 11 mg daily for men; pregnancy and lactation have different requirements. The adult tolerable upper intake level is 40 mg daily from all sources. Sustained excess zinc can impair copper absorption and contribute to hematological, neurological, or immune problems.

Biological Mechanism

GHK-Cu is a copper-binding tripeptide complex with experimental matrix-remodeling actions. Copper is a cofactor in connective-tissue synthesis, antioxidant defense, and mitochondrial

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energy metabolism; enhancement of those functions after GLOW administration is a mechanistic hypothesis. [1,37] GHK-Cu increased collagen synthesis in fibroblast cultures and altered collagen, glycosaminoglycan, decorin, and biglycan responses in rat wounds and fibroblasts. [2,3] In a mouse inflammatory lung-injury model, it reduced NF-κB-associated signaling and inflammatory mediators, including TNF-α and IL-6. [38] Reviews also describe elastin, cell-migration, and capillary-growth effects. [1] The frequently cited "over 4,000 genes" finding is an estimate from GHK-treated cultured-cell expression profiles, not a demonstrated whole-body regenerative program after injection. [30,39] These observations support a plausible remodeling rationale while keeping copper biology, cell responses, and clinical outcomes distinct.

BPC-157 is a 15-amino-acid experimental peptide with cytoprotective and repair-related findings in preclinical systems. [4,5,13] In rat tendon-derived cultures, it promoted outgrowth, migration, spreading, and survival under oxidative stress, associated with FAK–paxillin signaling; it did not directly increase proliferation in that experiment. [5] In human endothelial cells it increased VEGFR2 expression and internalization and activated VEGFR2–Akt–eNOS signaling, without increasing VEGF-A expression. Rat ischemic-limb experiments showed improved perfusion and vessel formation. [31] Gastrointestinal and inflammatory research discusses barrier preservation, cytokine modulation, and interactions with nitric oxide, dopamine, and serotonin systems. [4,40] FGF-2-related repair signaling and coordinated growth-factor effects remain mechanistic proposals rather than a demonstrated universal response to GLOW. [4,31] These pathways plausibly connect vascular restoration, inflammation control, and matrix remodeling to tissue recovery. [5,13,31]

Full-length TB4 is a 43-amino-acid peptide that binds and sequesters globular actin, regulating the monomer pool available for filament assembly. [6,17] TB4 improved wound repair and cardiac injury outcomes in particular animal models. [8,9] Corneal-cell studies showed reduced TNF-α-induced NF-κB activation and increased laminin-5 production through a pathway distinct from TGF-β signaling. [7,12] Its repair-related effects include migration and local vascular responses, with outcome and tissue context determining the interpretation. [10,11,23,24,27,28]

Synergies in the combination are plausible. [7-11,17]

References

[1] Pickart L, Margolina A. Regenerative and Protective Actions of the GHK-Cu Peptide in the Light of the New Gene Data. *Int J Mol Sci*. 2018;19(7):1987. doi:10.3390/ijms19071987

[2] Maquart FX, Pickart L, Laurent M, Gillery P, Monboisse JC, Borel JP. Stimulation of collagen synthesis in fibroblast cultures by the tripeptide-copper complex glycyl-L-histidyl-L-lysine-Cu2+. *FEBS Lett*. 1988;238(2):343–346. doi:10.1016/0014-5793(88)80509-x

[3] Siméon A, Wegrowski Y, Bontemps Y, Maquart FX. Expression of glycosaminoglycans and small proteoglycans in wounds: modulation by the tripeptide-copper complex glycyl-L-histidyl-L-lysine-Cu(2+). *J Invest Dermatol*. 2000;115(6):962–968. doi:10.1046/j.1523-1747.2000.00166.x

[4] Sikiric P, Seiwerth S, Rucman R, et al. Stable gastric pentadecapeptide BPC 157: novel therapy in gastrointestinal tract. *Curr Pharm Des*. 2011;17(16):1612–1632. doi:10.2174/138161211796196954

[5] Chang CH, Tsai WC, Lin MS, Hsu YH, Pang JH. The promoting effect of pentadecapeptide BPC 157 on tendon healing involves tendon outgrowth, cell survival, and cell migration. *J Appl Physiol (1985)*. 2011;110(3):774–780. doi:10.1152/japplphysiol.00945.2010

[6] Goldstein AL, Hannappel E, Sosne G, Kleinman HK. Thymosin β4: a multi-functional regenerative peptide. Basic properties and clinical applications. *Expert Opin Biol Ther*. 2012;12(1):37–51. doi:10.1517/14712598.2012.634793

[7] Sosne G, Qiu P, Christopherson PL, Wheater MK. Thymosin beta 4 suppression of corneal NFkappaB: a potential anti-inflammatory pathway. *Exp Eye Res*. 2007;84(4):663–669. doi:10.1016/j.exer.2006.12.004

[8] Malinda KM, Sidhu GS, Mani H, et al. Thymosin beta4 accelerates wound healing. *J Invest Dermatol*. 1999;113(3):364–368. doi:10.1046/j.1523-1747.1999.00708.x

[9] Bock-Marquette I, Saxena A, White MD, Dimaio JM, Srivastava D. Thymosin beta4 activates integrin-linked kinase and promotes cardiac cell migration, survival and cardiac repair. *Nature*. 2004;432(7016):466–472. doi:10.1038/nature03000

[10] Sosne G, Szliter EA, Barrett R, Kernacki KA, Kleinman H, Hazlett LD. Thymosin beta 4 promotes corneal wound healing and decreases inflammation in vivo following alkali injury. *Exp Eye Res*. 2002;74(2):293–299. doi:10.1006/exer.2001.1125

[11] Smart N, Risebro CA, Melville AA, et al. Thymosin beta4 induces adult epicardial progenitor mobilization and neovascularization. *Nature*. 2007;445(7124):177–182. doi:10.1038/nature05383

[12] Sosne G, Xu L, Prach L, et al. Thymosin beta 4 stimulates laminin-5 production independent of TGF-beta. *Exp Cell Res*. 2004;293(1):175–183. doi:10.1016/j.yexcr.2003.09.022

[13] Staresinic M, Sebecic B, Patrlj L, et al. Gastric pentadecapeptide BPC 157 accelerates healing of transected rat Achilles tendon and in vitro stimulates tendocytes growth. *J Orthop Res*. 2003;21(6):976–983. doi:10.1016/s0736-0266(03)00110-4

[14] Miller TR, Wagner JD, Baack BR, Eisbach KJ. Effects of topical copper tripeptide complex on CO2 laser-resurfaced skin. *Arch Facial Plast Surg*. 2006;8(4):252–259. doi:10.1001/archfaci.8.4.252

[15] Kang EA, Han YM, An JM, et al. BPC157 as Potential Agent Rescuing from Cancer Cachexia. *Curr Pharm Des*. 2018;24(18):1947–1956. doi:10.2174/1381612824666180614082950

[16] Xing Y, Ye Y, Zuo H, Li Y. Progress on the Function and Application of Thymosin β4. *Front Endocrinol (Lausanne)*. 2021;12:767785. doi:10.3389/fendo.2021.767785

[17] Xue B, Leyrat C, Grimes JM, Robinson RC. Structural basis of thymosin-β4/profilin exchange leading to actin filament polymerization. *Proc Natl Acad Sci U S A*. 2014;111(43):E4596–E4605. doi:10.1073/pnas.1412271111

[18] Ruff D, Crockford D, Girardi G, Zhang Y. A randomized, placebo-controlled, single and multiple dose study of intravenous thymosin beta4 in healthy volunteers. *Ann N Y Acad Sci*. 2010;1194:223–229. doi:10.1111/j.1749-6632.2010.05474.x

[19] Guarnera G, DeRosa A, Camerini R, 8 European sites. The effect of thymosin treatment of venous ulcers. *Ann N Y Acad Sci*. 2010;1194:207–212. doi:10.1111/j.1749-6632.2010.05490.x

[20] Sosne G, Ousler GW. Thymosin beta 4 ophthalmic solution for dry eye: a randomized, placebo-controlled, Phase II clinical trial conducted using the controlled adverse environment (CAE™) model. *Clin Ophthalmol*. 2015;9:877–884. doi:10.2147/opth.s80954

[21] Sosne G, Kleinman HK, Springs C, Gross RH, Sung J, Kang S. 0.1% RGN-259 (Thymosin ß4) Ophthalmic Solution Promotes Healing and Improves Comfort in Neurotrophic Keratopathy Patients in a Randomized, Placebo-Controlled, Double-Masked Phase III Clinical Trial. *Int J Mol Sci*. 2023;24(1):554. doi:10.3390/ijms24010554

[22] Philp D, Nguyen M, Scheremeta B, et al. Thymosin beta4 increases hair growth by activation of hair follicle stem cells. *FASEB J*. 2004;18(2):385–387. doi:10.1096/fj.03-0244fje

[23] Li X, Wang L, Chen C. Effects of exogenous thymosin β4 on carbon tetrachloride-induced liver injury and fibrosis. *Sci Rep*. 2017;7(1):5872. doi:10.1038/s41598-017-06318-5

[24] Kim J, Lee C, Han J, et al. Targeted Deletion of Thymosin Beta 4 in Hepatic Stellate Cells Ameliorates Liver Fibrosis in a Transgenic Mouse Model. *Cells*. 2023;12(12):1658. doi:10.3390/cells12121658

[25] Cha HJ, Jeong MJ, Kleinman HK. Role of thymosin beta4 in tumor metastasis and angiogenesis. *J Natl Cancer Inst*. 2003;95(22):1674–1680. doi:10.1093/jnci/djg100

[26] Caers J, Hose D, Kuipers I, et al. Thymosin β4 has tumor suppressive effects and its decreased expression results in poor prognosis and decreased survival in multiple myeloma. *Haematologica*. 2010;95(1):163–167. doi:10.3324/haematol.2009.006411

[27] Gonzalez-Franquesa A, Stocks B, Borg ML, et al. Discovery of thymosin β4 as a human exerkine and growth factor. *Am J Physiol Cell Physiol*. 2021;321(5):C770–C778. doi:10.1152/ajpcell.00263.2021

[28] Tokura Y, Nakayama Y, Fukada S, et al. Muscle injury-induced thymosin β4 acts as a chemoattractant for myoblasts. *J Biochem*. 2011;149(1):43–48. doi:10.1093/jb/mvq115

[29] Janoshik Analytical. TB4/TB-500/TB4(17-23) analysis. Accessed October 21, 2026. Janoshik

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[30] Pickart L, Vasquez-Soltero JM, Pickart FD, Majnarich J. GHK, the Human Skin Remodeling Peptide, Induces Anti-Cancer Expression of Numerous Caspase, Growth Regulatory, and DNA Repair Genes. *J Anal Oncol*. 2014;3(2):79–87. doi:10.6000/1927-7229.2014.03.02.2

[31] Hsieh MJ, Liu HT, Wang CN, et al. Therapeutic potential of pro-angiogenic BPC157 is associated with VEGFR2 activation and up-regulation. *J Mol Med (Berl)*. 2017;95(3):323–333. doi:10.1007/s00109-016-1488-y

[32] Lee E, Burgess K. Safety of Intravenous Infusion of BPC157 in Humans: A Pilot Study. *Altern Ther Health Med*. 2025;31(5):20–24. PubMed

[33] Lee E, Padgett B. Intra-Articular Injection of BPC 157 for Multiple Types of Knee Pain. *Altern Ther Health Med*. 2021;27(4):8–13. PubMed

[34] Wang X, Liu L, Qi L, et al. A first-in-human, randomized, double-blind, single- and multiple-dose, phase I study of recombinant human thymosin β4 in healthy Chinese volunteers. *J Cell Mol Med*. 2021;25(17):8222–8228. doi:10.1111/jcmm.16693

[35] US Food and Drug Administration. Certain Bulk Drug Substances for Use in Compounding that May Present Significant Safety Risks. Accessed October 21, 2026. FDA

[36] National Institutes of Health, Office of Dietary Supplements. Zinc: Fact Sheet for Health Professionals. Accessed October 21, 2026. NIH ODS

[37] National Institutes of Health, Office of Dietary Supplements. Copper: Fact Sheet for Health Professionals. Accessed October 21, 2026. NIH ODS

[38] Park JR, Lee H, Kim SI, Yang SR. The tri-peptide GHK-Cu complex ameliorates lipopolysaccharide-induced acute lung injury in mice. *Oncotarget*. 2016;7(36):58405–58417. doi:10.18632/oncotarget.11168

[39] Pickart L, Vasquez-Soltero JM, Margolina A. GHK and DNA: resetting the human genome to health. *Biomed Res Int*. 2014;2014:151479. doi:10.1155/2014/151479

[40] Park JM, Lee HJ, Sikiric P, Hahm KB. BPC 157 Rescued NSAID-cytotoxicity Via Stabilizing Intestinal Permeability and Enhancing Cytoprotection. *Curr Pharm Des*. 2020;26(25):2971–2981. doi:10.2174/1381612826666200523180301